17.1 Parameterized curves

A curve in $\mathbb{R}^2$ can be described by parametric equations where

$$x = f(t) \qquad \text{and} \qquad y = g(t)$$

where f and g are functions of the parameter t. If the curve is in $\mathbb{R}^3$, we can get parametric equations by adding a third equation $z = h(t)$.

→ Do problems 1-4.

Parametric equations can also be described using vector-valued functions as

$$\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$$

or

$$\vec{r}(t) = f(t)\vec{i} + g(t)\vec{j} + h(t)\vec{k}.$$

Tip: Think of these as position functions for a point moving along the curve.

The line through the point (x_0, y_0, z_0) and in the direction of $\vec{v} = \langle a, b, c \rangle$ has parameterization

$$\vec{r}(t) = \langle x_0, y_0, z_0 \rangle + \vec{v}t = \langle x_0 + at, y_0 + bt, z_0 + ct \rangle.$$

→ Do problems 5-8.